# Experiment to understand feature scaling.

In [1]:
```
import numpy as np
```

In [2]:
```
array=np.random.randint(1,100,16) # randomly generate 16 numbers between 1 t
array
```

Out[2]:
```
array([95,  3, 59, 89, 77, 95, 58, 56, 42, 10, 92, 43, 91,  5, 74, 30])
```

In [3]:
```
array.mean()
```

Out[3]:
```
np.float64(57.4375)
```

In [4]:
```
np.percentile(array,25)
```

Out[4]:
```
np.float64(39.0)
```

In [5]:
```
np.percentile(array,50)
```

Out[5]:
```
np.float64(58.5)
```

In [6]:
```
np.percentile(array,75)
```

Out[6]:
```
np.float64(89.5)
```

In [7]:
```
np.percentile(array,100)
```

Out[7]:
```
np.float64(95.0)
```

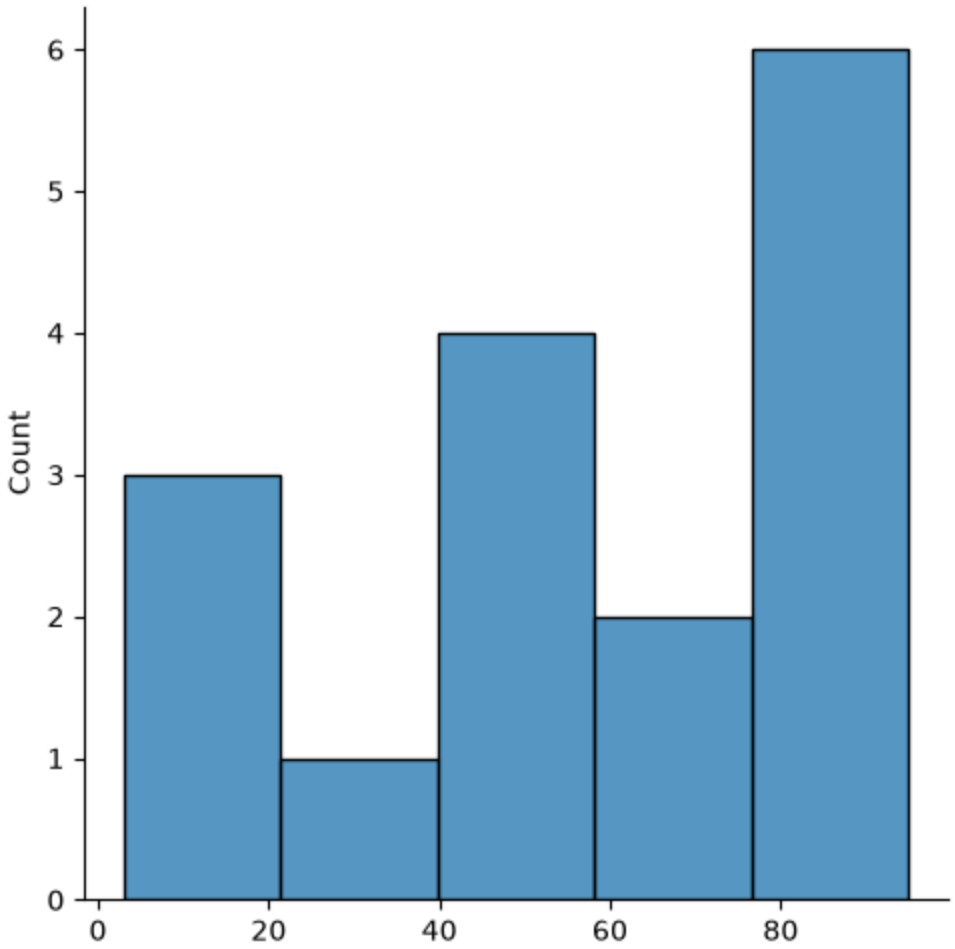
In [13]:
```
#outliers detection
def outDetection(array):
  sorted(array)
  Q1,Q3=np.percentile(array,[25,75])
  IQR=Q3-Q1
  lr=Q1-(1.5*IQR)
  ur=Q3+(1.5*IQR)
  return lr,ur
```

In [17]:
```
lr,ur=outDetection(array)
lr,ur
```

Out[17]:
```
(np.float64(-36.75), np.float64(165.25))
```

In [18]:
```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

Out[18]:
```
<seaborn.axisgrid.FacetGrid at 0x16f81bc50>
```




In [19]:
```
sns.distplot(array)
```

```
/var/folders/_p/_l2hmmfs4gb27vbdzw9snc880000gn/T/ipykernel_48654/1133588802.
py:1: UserWarning:

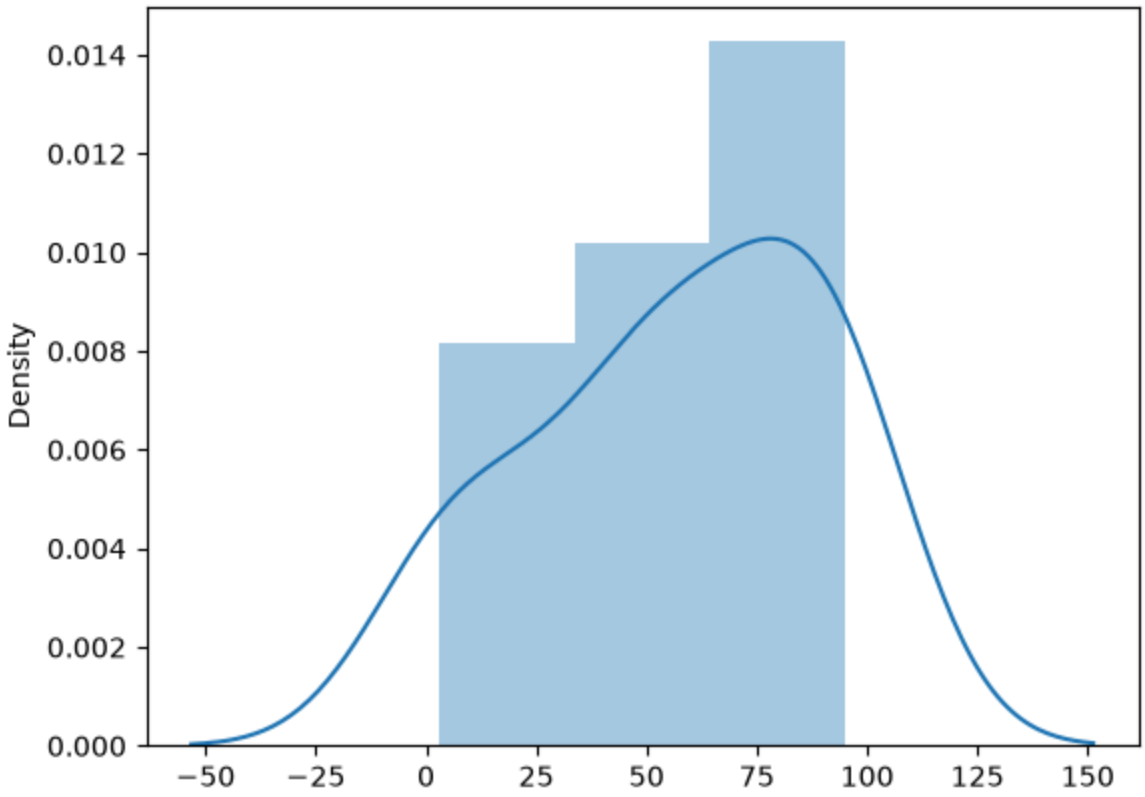
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(array)
```

Out[19]: <Axes: ylabel='Density'>

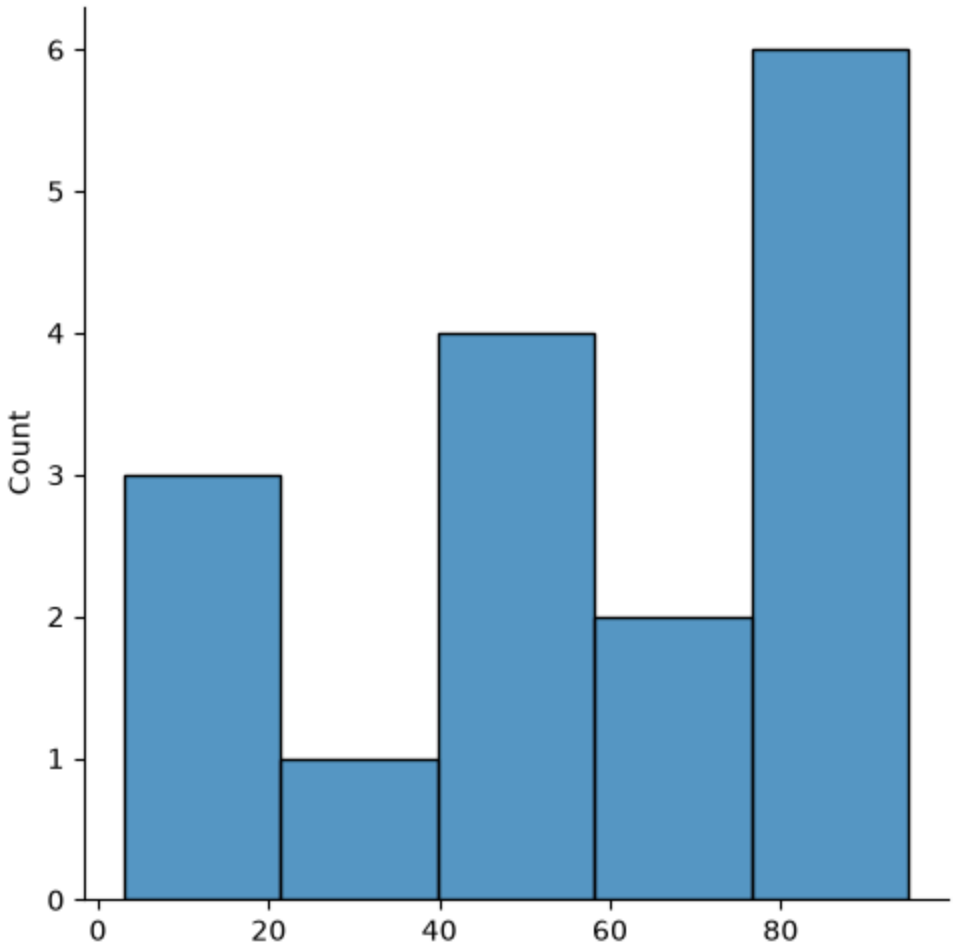



In [20]:
```
new_array=array[(array>lr) & (array<ur)]
new_array
```

Out[20]:
```
array([95,  3, 59, 89, 77, 95, 58, 56, 42, 10, 92, 43, 91,  5, 74, 30])
```

In [22]:
```
sns.displot(new_array)
```

Out[22]:
```
<seaborn.axisgrid.FacetGrid at 0x16fc3da90>
```




In [23]:
```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

Out[23]:
```
(np.float64(-36.75), np.float64(165.25))
```

In [24]:
```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

Out[24]:
```
array([95,  3, 59, 89, 77, 95, 58, 56, 42, 10, 92, 43, 91,  5, 74, 30])
```

In [25]:
```
sns.distplot(final_array)
```

```
/var/folders/_p/_l2hmmfs4gb27vbdzw9snc880000gn/T/ipykernel_48654/209491988.py:1: UserWarning:

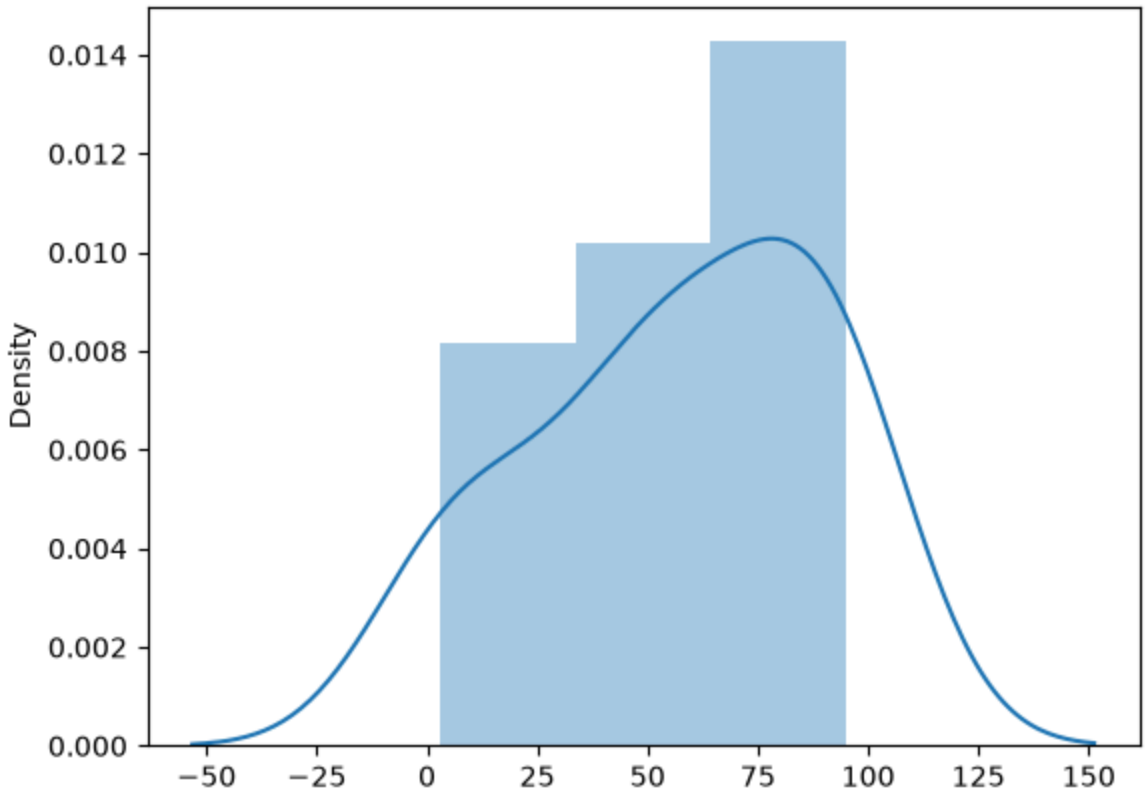
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(final_array)
```

Out[25]:
```
<Axes: ylabel='Density'>
```


0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000
Density
−50
−25
0
25
50
75
100
125
150

In [ ]: